

Cosine Similarity

Do vectors kitne 'same direction' mein hain, magnitude ignore karke

01 Opening Hook

STORY TIME

Socho do log alag alag speed se same direction mein chal rahe hain – ek slow, ek fast, lekin dono North ki taraf. Unki speed alag hai lekin 'intent' (direction) same hai. AI mein bhi hum kabhi 'kitna similar' yeh dekhna chahte hain sirf direction ke basis par, size ignore karke – iska naam hai **Cosine Similarity**.

02 Why Does This Concept Exist?

THE PROBLEM

Text embeddings mein, kisi document ki 'length' uske meaning ko change nahi karni chahiye – ek chhota aur ek bada document same topic pe ho sakte hain. Cosine similarity invent hua taaki hum sirf 'direction' (meaning) compare kar sakein, magnitude (length/intensity) ko ignore karte hue.

03 Explain Like I'm 10

SIMPLE WORDS

Socho do arrows hain jo almost same direction mein point kar rahe hain, lekin ek chhota hai aur ek bada. Cosine similarity kehta hai 'inka angle kitna chhota hai' – agar angle 0 hai (same direction), similarity = 1 (perfect match). Agar angle 90 degree hai (perpendicular), similarity = 0 (unrelated).

04 Build Intuition First (No Maths Yet)

FEEL IT FIRST

Cosine similarity do vectors ke beech ke angle ka cosine calculate karta hai. Yeh -1 (opposite) se 1 (identical direction) tak range karta hai. Text/embedding search mein yeh sabse popular similarity metric hai kyunki yeh document length ke effect ko naturalized kar deta hai.

05 Ab Thoda Maths

FORMULA TIME

Formula:

$\text{cosine_similarity}(A,B) = (A \cdot B) / (\|A\| * \|B\|)$ - dot product divide by dono vectors ke magnitudes ka product.

06 Visualize Everything

PICTURE IT

Similar direction:

A B
 \ |
 \ | angle~0
cos~1 (very similar)

Perpendicular (unrelated):

A
 |
 |____ B angle=90
cos=0 (unrelated)

07 AI Connection

WHERE AI USES THIS

Semantic Search / RAG: Jab tum kisi query se related documents dhoondte ho, query embedding aur document embeddings ke beech cosine similarity calculate karke sabse close matches nikalte ho.

Recommendation Systems: User preference vectors aur item vectors ke beech cosine similarity se personalized recommendations banti hain.

08 Real Industry Example

REAL COMPANIES

OpenAI/Anthropic (Embeddings API): Semantic search applications cosine similarity ko standard metric ke roop mein use karte hain embeddings compare karne ke liye.

Pinecone/Weaviate (Vector DBs): Vector databases cosine similarity search ko highly optimize karte hain millions of vectors ke beech fast retrieval ke liye.

09 Explain Like I'm Going To Build It

HOW YOU'D CODE IT

PyTorch: `torch.nn.functional.cosine_similarity(a, b, dim=0)`. RAG pipelines mein: query ko embed karo, document embeddings se cosine similarity calculate karo, top-k matches return karo.

10 Common Misconceptions

BEGINNER MISTAKES

Galat soch: Cosine similarity aur Euclidean distance same cheez hain.

Sahi baat: Yeh different hain - cosine sirf direction dekhta hai (magnitude-independent), Euclidean distance actual 'gap' measure karta hai jisme magnitude bhi matter karta hai.

Galat soch: Cosine similarity hamesha 0 se 1 ke beech hoti hai.

Sahi baat: Technically yeh -1 se 1 tak range kar sakti hai, lekin text embeddings mein (jo usually non-negative components rakhte hain) practically 0 se 1 ke beech dikhti hai.

11 Interview Questions

INTERVIEW CORNER

[Beginner] Q: Cosine similarity kya measure karti hai?

A: Do vectors ke beech ka angle - woh kitni 'same direction' mein hain, magnitude ko ignore karke.

[Intermediate] Q: RAG systems mein cosine similarity kyun use hoti hai Euclidean distance ki jagah?

A: Kyunki text ki length embedding ki magnitude ko affect karti hai, lekin meaning (direction) usse independent hoti hai - cosine similarity is length-bias ko naturalized kar deta hai.

[Advanced] Q: High-dimensional embeddings mein cosine similarity search efficiently kaise ki jaati hai?

A: Approximate Nearest Neighbor (ANN) algorithms (jaise HNSW) use hote hain jo brute-force comparison ki jagah efficient indexing structures banate hain, jisse millions of vectors mein bhi fast search ho paati hai.

12 Exercises

PRACTICE TIME

[Easy] Agar do vectors exactly same direction mein hain, unka cosine similarity kya hoga?

Solution: 1.

[Medium] Agar do vectors perpendicular hain, cosine similarity kya hogi?

Solution: 0.

[Hard] $A=[1,0]$ aur $B=[100,0]$ - inka cosine similarity kya hoga aur kyun?

Solution: 1, kyunki dono same direction mein hain - magnitude ka farak (1 vs 100) cosine similarity ko affect nahi karta, sirf direction matter karta hai.

13 Golden Summary

Concept	Easy Meaning
Cosine Similarity	Vectors ke beech angle-based similarity (-1 se 1)
Formula	$(A \cdot B) / (A * B)$

RAG/Search Use	Semantic search ka standard metric
Magnitude-independent	Sirf direction compare karta hai
Range	1 = identical direction, 0 = unrelated, -1 = opposite

14 One-Line Memory Trick

“Cosine similarity dekhta hai 'kis taraf ja rahe ho', 'kitni door ja rahe ho' nahi.”